

# Network Protocols

Network protocols are agreed rules that allow computers, phones, servers, routers, and other networked devices to communicate. A protocol defines how information is formatted, addressed, transmitted, received, checked, and interpreted. Without common protocols, devices made by different manufacturers would have difficulty communicating reliably.

This document provides a simple introduction to important network protocols. It covers the basic ideas behind communication protocols and then explains common protocols used at different layers of computer networks.

# 1. Introduction to Network Protocols

A network protocol is a set of rules used during communication between two or more devices. These rules can describe message formats, timing, addressing, error handling, authentication, and connection management.

Protocols make communication predictable. For example, when a web browser requests a webpage, several protocols work together. DNS may translate a domain name into an IP address, TCP may provide reliable transport, IP may route packets, and HTTP or HTTPS may define the web request itself.

Protocols are usually designed to solve particular communication problems. Some focus on reliable delivery, some on addressing and routing, and others on applications such as email, file transfer, or web browsing.

## 2. Network Communication Models

Network communication is commonly explained using layered models. The OSI model divides networking into seven layers: physical, data link, network, transport, session, presentation, and application. The TCP/IP model is a practical model commonly described using link, internet, transport, and application layers.

Layering makes network design easier because each layer has a specific responsibility. A protocol at one layer can work with protocols at another layer without every protocol needing to understand all lower-level details.

For example, HTTP operates at the application level, TCP operates at the transport level, and IP operates at the internet or network level. Together they provide end-to-end communication.

### **3. Ethernet**

Ethernet is a major technology and protocol family used for local area networks. It defines how devices communicate over wired network connections and includes rules for frames, addressing, and physical transmission.

Ethernet frames contain source and destination MAC addresses along with a payload and error-detection information. Switches examine MAC addresses to forward frames toward the correct device.

Modern Ethernet is widely used in homes, offices, schools, and data centers. It can support many different speeds, from older low-speed implementations to very high-speed data-center links.

## 4. ARP

Address Resolution Protocol, or ARP, is used in IPv4 networks to discover the MAC address associated with a local IPv4 address. A device may know the destination IP address but still need the destination MAC address before sending an Ethernet frame.

The device broadcasts an ARP request asking which device owns a particular IP address. The device with that address responds with its MAC address. The result can be stored temporarily in an ARP cache.

ARP is useful inside local networks. It does not normally resolve addresses across the entire internet because routers separate different local network segments.

## 5. IP and IPv4

Internet Protocol provides logical addressing and packet delivery between networks. IPv4 uses 32-bit addresses, commonly written as four decimal numbers separated by dots.

An IPv4 packet contains source and destination addresses and other control information. Routers examine the destination address and use routing information to select a path toward the destination.

IPv4 addresses are divided into network and host portions using subnet masks or prefix lengths. Subnetting allows organizations to divide networks into smaller logical networks, improving address management and traffic control.

## 6. IPv6

IPv6 was developed to provide a much larger address space than IPv4. IPv6 addresses are 128 bits long and are normally written using hexadecimal notation separated by colons.

IPv6 also improves several aspects of network design, including address configuration and packet handling. Devices can use mechanisms such as Neighbor Discovery rather than ARP.

IPv6 supports global addressing for a very large number of devices. Although IPv4 remains widely used, IPv6 deployment continues as networks need more addresses and modern networking capabilities.

## 7. TCP

Transmission Control Protocol provides reliable, ordered, connection-oriented delivery of data. It is commonly used by applications that need accurate delivery rather than simply fast delivery.

TCP establishes a connection using a handshake and uses sequence numbers, acknowledgements, retransmission, and flow control. These mechanisms help detect missing or damaged data and allow it to be sent again.

TCP is widely used with HTTP and HTTPS, email protocols, file transfer, and many other applications. Its reliability introduces overhead, so it is not ideal for every type of network traffic.

## 8. UDP

User Datagram Protocol is a connectionless transport protocol. It provides a simple way to send datagrams without establishing a traditional TCP-style connection.

UDP does not guarantee delivery, ordering, or retransmission. This makes it lightweight and useful when speed and low delay are more important than perfect delivery.

UDP is commonly used for DNS queries, streaming-related applications, online games, voice communication, and other systems where an application can tolerate some packet loss or implement its own reliability mechanisms.

## 9. HTTP

Hypertext Transfer Protocol is the main protocol used for communication between web clients and web servers. A client such as a browser sends an HTTP request and a server returns an HTTP response.

HTTP defines methods such as GET, POST, PUT, PATCH, and DELETE. Responses include status codes such as 200 for success, 404 for a missing resource, and 500 for a server-side error.

HTTP is an application-layer protocol. In modern web communication, HTTP is commonly carried over secure connections using HTTPS.

## 10. HTTPS

HTTPS is HTTP protected using Transport Layer Security, commonly called TLS. It provides encryption, integrity protection, and authentication for web communication.

When a browser connects securely to a website, the TLS process helps establish cryptographic keys and verifies the server using a digital certificate. After the secure session is established, HTTP data can be transmitted confidentially.

HTTPS is important for protecting passwords, payment information, personal data, and general browsing activity from interception or modification during transmission.

## 11. DNS

Domain Name System translates human-readable domain names into information used by network applications, especially IP addresses. People can remember names such as example.com more easily than numerical addresses.

DNS uses a distributed hierarchy of servers. A resolver may contact DNS servers to obtain the required information and can cache results for a period defined by the record's TTL.

DNS commonly uses UDP for ordinary queries, while TCP can be used for larger responses or particular DNS operations. Modern DNS security and privacy technologies can add further protection.

## 12. DHCP

Dynamic Host Configuration Protocol automatically provides network configuration information to devices. A DHCP server can assign an IP address, subnet information, default gateway, DNS servers, and other options.

A typical DHCP exchange is often described using Discover, Offer, Request, and Acknowledgement messages. This allows a device to join a network without manually configuring every setting.

DHCP is common in home and business networks. It simplifies administration and reduces configuration errors, especially when many devices frequently connect and disconnect.

## **13. FTP and SFTP**

File Transfer Protocol, or FTP, is designed for transferring files between systems. Traditional FTP uses separate control and data connections and does not encrypt information by default.

SFTP is different from traditional FTP. It is a file-transfer protocol that operates over SSH and provides encrypted communication. Because of this, SFTP is often preferred when secure file transfer is required.

File-transfer systems should use appropriate authentication, permissions, encryption, and logging to reduce the risk of unauthorized access or exposure of sensitive files.

## **14. SMTP**

Simple Mail Transfer Protocol is primarily used to send email between mail servers and from email clients to mail submission servers.

SMTP uses commands and responses to establish a mail transaction. The sending system identifies itself, specifies the sender and recipients, and transfers the message content.

SMTP is mainly a sending protocol. Email retrieval is commonly handled by protocols such as IMAP or POP3. Modern email systems also use encryption and authentication to reduce abuse and protect communications.

## **15. IMAP and POP3**

Internet Message Access Protocol and Post Office Protocol version 3 are used to retrieve email.

IMAP is designed for accessing messages stored on a mail server. It supports folders, synchronization, and access from multiple devices. This makes it suitable for modern email use.

POP3 is simpler and traditionally focuses on downloading messages from a server to a client. Depending on configuration, messages may be removed from the server after download.

Both protocols can be secured using encrypted connections.

## 16. SSH

Secure Shell is a protocol used for secure remote access and administration. It provides encrypted communication between a client and a server.

SSH can be used for remote command-line sessions, secure file transfers, tunneling, and other administrative tasks. Authentication can use passwords or cryptographic keys, with key-based authentication commonly preferred for automated systems.

Because SSH provides powerful access, accounts should use strong authentication, appropriate permissions, and careful access controls.

## 17. SNMP

Simple Network Management Protocol is used to monitor and manage network devices such as routers, switches, servers, and printers.

SNMP can allow a management system to retrieve information such as interface status, traffic counters, memory use, and device health. It can also support notifications when particular events occur.

Different versions of SNMP provide different security capabilities. SNMPv3 adds stronger authentication and privacy features and is preferred when secure management communication is required.

## 18. ICMP

Internet Control Message Protocol is used for network control, diagnostics, and error reporting. It works alongside IP rather than serving as a general application data protocol.

Utilities such as ping commonly use ICMP Echo Request and Echo Reply messages to test reachability. Other ICMP messages can indicate conditions such as unreachable destinations or expired packet lifetimes.

ICMP is important for troubleshooting and for normal operation of IP networks. Network security policies may restrict some ICMP traffic, but blocking it indiscriminately can cause diagnostic and connectivity problems.

## 19. Routing Protocols

Routing protocols help routers exchange information about network paths. Examples include RIP, OSPF, EIGRP, and BGP.

RIP is relatively simple and uses hop count as its primary metric. OSPF is a link-state protocol commonly used inside organizations. BGP is the major routing protocol used to exchange routing information between autonomous systems on the internet.

Routing protocols allow networks to adapt when paths change. They are fundamental to scalable communication across interconnected networks.

## 20. Network Security Protocols

Network security protocols protect data and communication from interception, modification, impersonation, and unauthorized access.

TLS protects many application connections, including HTTPS. IPsec can provide security at the IP layer and is commonly associated with VPN technologies. SSH protects remote administration and related communications.

Security also depends on correct configuration. Strong protocols cannot fully protect a system if credentials are weak, certificates are mishandled, software is outdated, or access permissions are overly broad.

## 21. How Protocols Work Together

Network communication normally involves multiple protocols working as a stack. When a user opens a website, DNS may resolve the domain name, TCP or another transport mechanism may carry application data, IP provides addressing and routing, and Ethernet or Wi-Fi handles local transmission.

Each layer adds information needed by the corresponding protocol. At the receiving side, the process is reversed as the data moves upward through the stack.

Understanding protocol interaction is useful when troubleshooting. A failure may occur at the physical layer, local network layer, IP routing layer, transport layer, DNS service, or application server.

## 22. Summary

Network protocols are the foundation of digital communication. Ethernet and Wi-Fi connect devices locally, ARP and Neighbor Discovery help with local addressing, IP provides logical addressing and routing, and TCP or UDP provides transport services.

At the application level, protocols such as HTTP, DNS, DHCP, SMTP, IMAP, SSH, and SNMP support common network services. Security protocols such as TLS and IPsec add protection where confidentiality, integrity, and authentication are required.

Learning protocols as a layered system makes networking easier to understand. Instead of viewing a network as one complicated process, each protocol can be studied according to its specific purpose and then understood as part of the larger communication stack.